

Akash Agrawal

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EDUCATION

M.S. Computer Science | Arizona State University, Tempe, AZ Dec 2025

Courses: Statistical Machine Learning, Cloud Computing, Data Processing at Scale, Software Security, Semantic Web Mining

B.S. Computer Science | Nagpur University, IN Jul 2021

Courses: Data Structure & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, AI/ML

PROFESSIONAL EXPERIENCE

Software Engineer - AI Feb 2026 - present
NewsGenie, Inc. Mesa, AZ

- Built a real-time content protection service for publishers to encrypt and revoke access to digital content against unauthorized bots and scrapers, using Fast API, React, MongoDB, Redis, and AWS KMS.
- Designed per-user, per-article access control with sub-second revocation, blocking 7 distinct bot categories for publisher customers across service boundaries.
- Built an agentic content pipeline using LangChain with multi-LLM orchestration and automated review gates, cutting manual workflow by 1,000+ hours annually.

AI Engineer Aug 2024 - Jul 2025
Arizona State University Tempe, AZ

- Architected a deep learning pipeline on AWS Lambda and S3 to reduce data analysis time by 95% (40hr to 2hr per cycle).
- Engineered a multi-object tracking system using FairMOT and YOLOv8 to process 3500+ entities at 0.82 mAP.
- Standardized deployment environments using Docker and Terraform to eliminate config drift and cut setup time by 90%.
- Implemented system observability using AWS CloudWatch to ensure 99.9% uptime and accelerate incident resolution.

Software Engineer Jan 2023 - Nov 2023
LTI Mindtree (Client: CITI Bank) Remote

- Developed secure, scalable middleware using Node.js and Angular to automate config management for financial systems.
- Accelerated data processing by 35% through Oracle PL/SQL optimization, RESTful API integration, and parallel execution logic.
- Hardened application security by implementing SSO, RBAC, and full audit logging to protect sensitive financial data.
- Deployed an end-to-end CI/CD pipeline using Jenkins, Docker, Kubernetes, and SonarQube with enforced code reviews.

Software Engineer - Data Aug 2021 - Dec 2022
LTI Mindtree (Client: Avery Dennison) Remote

- Engineered an enterprise MDM pipeline using Informatica (ETL) to consolidate 10M global customer records for unified CRM.
- Boosted legacy pipeline efficiency by 30% through data enrichment using Python, Oracle PL/SQL, and REST APIs.
- Enhanced data reliability using automated validation testing and deduplication logic to improve governance metrics by 4x.
- Delivered 2.5M golden customer records to downstream systems to enable AI-driven analytics across Oracle CRM/CX.

PROJECTS

NL2SQL AI Agent | Python, RAG, HuggingFace, LangChain, LLM, ChromaDB, SQLite, PostgreSQL Summer 2025

- Reduced client query resolution time from 3 days to under 1hr by building a natural language interface to relational database.
- Built a reflection-based AI agent with 3 fault-tolerant retry loops using LangGraph, LangChain and RAG.
- Achieved 0.71 retrieval precision and reduced hallucinations by 60% through RAGAS evaluation and prompt engineering.

LambdaLens | AWS Lambda, OpenCV, Docker, FFmpeg, ResNet-34, MongoDB Fall 2024

- Deployed a scalable event-driven video analysis pipeline using AWS Lambda to process concurrent media streams in real time.
- Optimized Docker builds to cut cloud costs by 50% and latency, while ensuring observability via standardized CloudWatch logs.
- Achieved 90%+ inference accuracy on unseen video data by integrating ResNet-34 within a serverless environment.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, SQL, C#, C/C++, Bash

Frameworks: FastAPI, React, Node.js, Next.js, GraphQL, RESTful API, Spring Boot, .NET, Angular, Streamlit

Tools & Cloud: AWS (Lambda, S3, KMS, CloudWatch), Docker, Kubernetes, Github, JIRA, Postman, Informatica, Cursor, Claude

Databases: SQL/RDBMS - MySQL, PostgreSQL, Oracle DB, SQLite | NoSQL - MongoDB | Cache - Redis | Vector - ChromaDB

ACHIEVEMENTS

- Ranked 239th internationally out of 10,000 teams in the ICPC 2019.
- Awarded 'Super Crew' honor by LTI Mindtree for technical excellence and collaboration.
- Active member of SoDA, HackerDevils, GDSC at ASU and Github open-source community.